

The Stolen Szechuan Sauce

Forensic analysis of a confirmed intrusion at Galactic Condiments Inc., covering disk, memory, registry and network evidence recovered from the domain controller.

CASE	The Stolen Szechuan Sauce
SYSTEM ANALYZED	DC01 — Domain Controller
ANALYST	Marco Helie — independent analysis (solo)
REPORT DATE	23 July 2025

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01 EXECUTIVE SUMMARY

Galactic Condiments Inc. experienced a confirmed data breach resulting in the theft of their confidential “Szechuan Sauce 2.0” formula. A suspicious file was discovered on a dark web marketplace, prompting a digital forensic investigation. I was hired to analyze forensic artifacts from the company’s domain controller (DC01) and related network traffic to determine the root cause, entry vector, and scope of the compromise.

Using tools such as Volatility, FTK Imager / Autopsy, Registry Explorer, Wireshark, and VirusTotal, I identified that the attacker gained access through brute-forcing Remote Desktop Protocol (RDP). The attacker deployed malware (“coreupdater.exe”) which provided remote shell access and installed persistence mechanisms via Windows services and registry keys.

Artifacts confirm that the malware communicated with an external command-and-control (C2) server and was moved across folders, indicating post-exploitation activity. The malware originated from an IP address with a known history of malicious activity.

This report provides a step-by-step breakdown of the forensic methods used to uncover these findings and includes annotated evidence. While the immediate compromise was limited to DC01, the malware’s capabilities and persistence mechanisms suggest the potential for lateral movement, making containment and remediation critical.

02 TOOLS & METHODOLOGY

To investigate the breach, I used multiple industry-standard digital forensic tools to analyze memory dumps, disk images, and network captures. The goal was to extract evidence of compromise, determine the timeline of the attack, and identify any persistence mechanisms or indicators of compromise (IOCs). Below is a summary of the tools used and their role in the investigation.

FTK Imager (AccessData, 2022)

FTK Imager was used to mount and explore the .E01 disk image of the DC01 system. I used it to locate key files and export registry hives (e.g., SYSTEM and SOFTWARE) for further analysis.

Autopsy (Basis Technology, 2024)

Autopsy was used as a graphical interface to further examine the contents of the DC01 disk image. It allowed for efficient timeline analysis, keyword searches, and viewing of deleted files or suspicious artifacts. I used it to cross-reference malware-related file activity and confirm metadata such as file creation / modification times.

Registry Explorer (Zimmerman Tools, 2023)

To analyze exported registry hives from FTK Imager, I used Eric Zimmerman’s Registry Explorer. This allowed me to extract timezone settings, installed services, and evidence of persistence.

Volatility 2.6 (Volatility Foundation, 2024)

Volatility was used to analyze the memory dump from DC01. It helped identify running processes, suspicious executables (e.g., coreupdater.exe), and memory-resident malware behavior.

Wireshark (Wireshark Foundation, 2024)

Wireshark was used to examine the Case001.pcap network traffic capture. I searched for suspicious traffic, including HTTP GET requests, command-and-control (C2) communication, and external IPs.

VirusTotal & Hybrid Analysis (VirusTotal, n.d.)

These platforms were used to validate the malicious nature of identified files and IP addresses. I cross-checked hash values and URLs against known malware databases.

03 SYSTEM ARTIFACT ANALYSIS: DC01

Q1 What is the operating system of the server?

ANSWER

Windows Server 2012 R2 Standard (Evaluation)

TOOL USED

FTK Imager (registry / system files), Autopsy

METHOD

Identified through the registry key `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion`

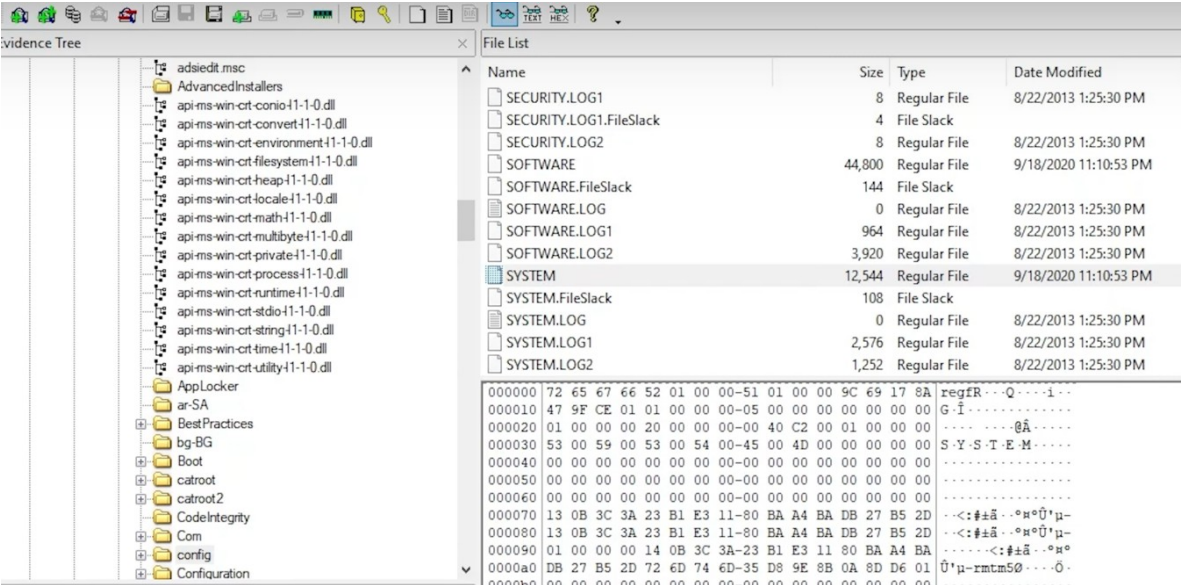


EXHIBIT 1 SYSTEM and SOFTWARE registry hives located in the mounted .E01 image

{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	C:\Windows	Current location of %SystemRoot%
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	Windows User	Current registered owner
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion		Current registered organization
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	Windows Server 2012 R2 Standard Evaluation	Current OS name
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	2020-09-17 16:43:59.0000000	Current OS install date
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	Server	Current OS installation type
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	ServerStandardEval	Current OS version and install int
{00000000-0000-0000-0000-000000000000}\Microsoft\Windows NT\CurrentVersion	9600	Current OS version and install int

EXHIBIT 2 CurrentVersion registry values confirming the OS name and install date

Q2 What is the operating system of the desktop?

SCOPE NOTE

Not analyzed. This report focuses solely on DC01.

Q3 What was the local time of the server?

ANSWER

Pacific Standard Time

TOOL USED

Registry Explorer, Autopsy

METHOD

Found in the control set registry under `TimeZoneInformation`

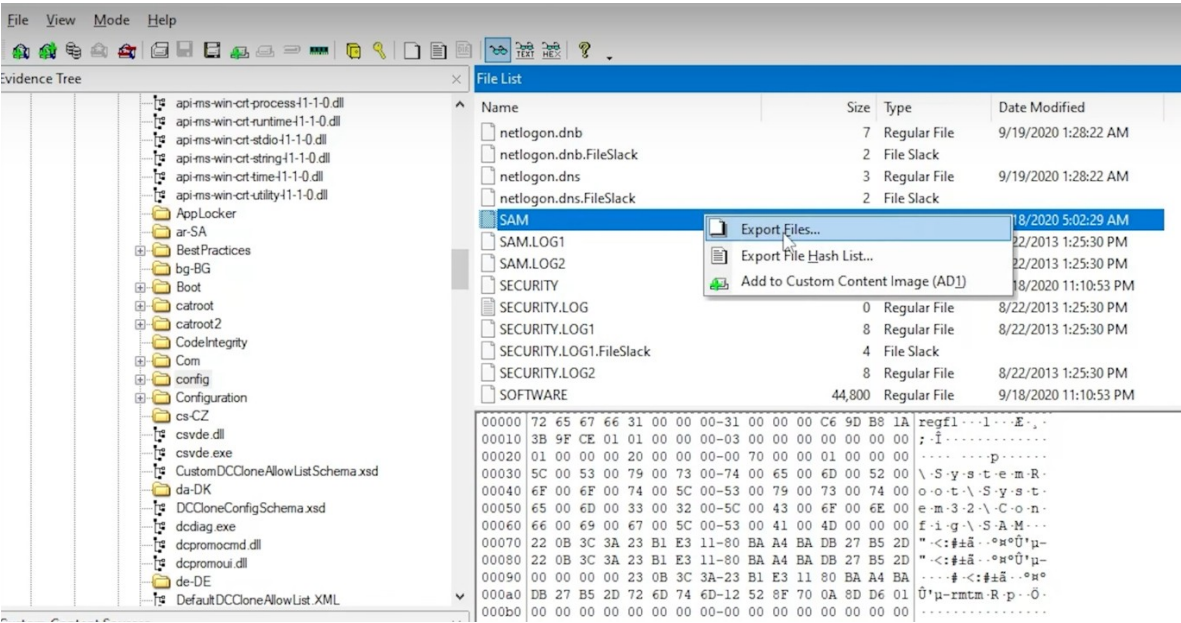


EXHIBIT 3 Exporting the registry hives from the mounted image in FTK Imager

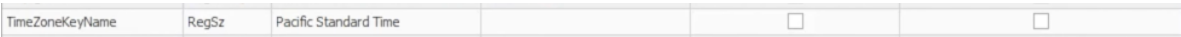


EXHIBIT 4 TimeZoneKeyName value resolving to Pacific Standard Time

Q4 Was there a breach?

ANSWER

Yes.

Q5 What was the initial entry vector?

Answer: The attacker gained access to the DC01 system via a brute-force attack against Remote Desktop Protocol (RDP). This conclusion is supported by a sequence of failed login attempts (Windows Event ID 4625) from the external IP address 194.61.24.102, followed by a successful login (Event ID 4624) shortly after.

Using **FTK Imager / Autopsy**, I mounted the DC01 .E01 disk image and exported the **Security.evtx** log. Then, I reviewed the log using **Event Log Explorer** to trace the authentication activity. The repeated 4625 events indicated failed login attempts via RDP. These were followed by a successful logon event, suggesting that the attacker eventually guessed or cracked a valid password.

This is a common attack vector, particularly when weak or default credentials are used and RDP is exposed to the internet without proper network segmentation or multi-factor authentication (MFA) (CISA, 2021).

“Brute-force attacks on exposed RDP ports continue to be a top entry vector for threat actors.”

— CISA, 2021

TOOL USED

Wireshark, FTK Imager / Autopsy, Registry Explorer

ie	Tag	Record Number	Event Record Id	Computer	Time Created	Event Id	Remote Host	Payload Data1
3856		7411	7411	CITADEL-DC01.C137.local	2020-09-19 03:21:30	4625	kali (-)	Target: \Administrator
3855		7410	7410	CITADEL-DC01.C137.local	2020-09-19 03:21:30	4625	kali (-)	Target: \Administrator
3854		7409	7409	CITADEL-DC01.C137.local	2020-09-19 03:21:30	4625	kali (-)	Target: \Administrator
3853		7408	7408	CITADEL-DC01.C137.local	2020-09-19 03:21:29	4625	kali (-)	Target: \Administrator
3852		7407	7407	CITADEL-DC01.C137.local	2020-09-19 03:21:29	4625	kali (-)	Target: \Administrator
3851		7406	7406	CITADEL-DC01.C137.local	2020-09-19 03:21:29	4625	kali (-)	Target: \Administrator
3850		7405	7405	CITADEL-DC01.C137.local	2020-09-19 03:21:29	4625	kali (-)	Target: \Administrator
3849		7404	7404	CITADEL-DC01.C137.local	2020-09-19 03:21:29	4625	kali (-)	Target: \Administrator
3848		7403	7403	CITADEL-DC01.C137.local	2020-09-19 03:21:28	4625	kali (-)	Target: \Administrator
3847		7402	7402	CITADEL-DC01.C137.local	2020-09-19 03:21:28	4625	kali (-)	Target: \Administrator
3846		7401	7401	CITADEL-DC01.C137.local	2020-09-19 03:21:28	4625	kali (-)	Target: \Administrator
3845		7400	7400	CITADEL-DC01.C137.local	2020-09-19 03:21:28	4625	kali (-)	Target: \Administrator
3844		7399	7399	CITADEL-DC01.C137.local	2020-09-19 03:21:27	4625	kali (-)	Target: \Administrator
3843		7398	7398	CITADEL-DC01.C137.local	2020-09-19 03:21:27	4625	kali (-)	Target: \Administrator
3842		7397	7397	CITADEL-DC01.C137.local	2020-09-19 03:21:27	4625	kali (-)	Target: \Administrator
3841		7396	7396	CITADEL-DC01.C137.local	2020-09-19 03:21:27	4625	kali (-)	Target: \Administrator
3840		7395	7395	CITADEL-DC01.C137.local	2020-09-19 03:21:27	4625	kali (-)	Target: \Administrator
3839		7394	7394	CITADEL-DC01.C137.local	2020-09-19 03:21:26	4625	kali (-)	Target: \Administrator
3838		7393	7393	CITADEL-DC01.C137.local	2020-09-19 03:21:26	4625	kali (-)	Target: \Administrator

EXHIBIT 5 Repeated Event ID 4625 failures against the Administrator account

CITADEL-DC01.C137.local	2020-09-19 03:22:37	4624	CITADEL-DC01 (194.61.24.102)	Target: C137\Administrator
CITADEL-DC01.C137.local	2020-09-19 03:22:36	4624	kali (-)	Target: C137\Administrator
CITADEL-DC01.C137.local	2020-09-19 03:22:09	4624	CITADEL-DC01 (194.61.24.102)	Target: C137\Administrator
CITADEL-DC01.C137.local	2020-09-19 03:22:07	4624	kali (-)	Target: C137\Administrator
CITADEL-DC01.C137.local	2020-09-19 03:21:48	4624	CITADEL-DC01 (194.61.24.102)	Target: C137\Administrator
CITADEL-DC01.C137.local	2020-09-19 03:21:46	4624	kali (-)	Target: C137\Administrator

EXHIBIT 6 Event ID 4624 successful logons originating from the attacker IP

84319	2020-09-19 02:19:13.414319	194.61.24.102	10.42.85.10	ICMP	42 Echo (ping) request id=0xef6f, seq=0/0, ttl=56 (reply in 84325)
84320	2020-09-19 02:19:13.414353	194.61.24.102	10.42.85.10	TCP	58 64385 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
84321	2020-09-19 02:19:13.414370	194.61.24.102	10.42.85.10	TCP	54 64385 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
84322	2020-09-19 02:19:13.414386	194.61.24.102	10.42.85.10	ICMP	54 Timestamp request id=0xe076, seq=0/0, ttl=51
84323	2020-09-19 02:19:13.414685	Vmware_el:84:e6	Broadcast	ARP	60 Who has 10.42.85.100? Tell 10.42.85.10
84324	2020-09-19 02:19:13.414690	Vmware_el:84:e6	ARP	42 10.42.85.100 is at 00:0c:29:95:cd:21	
84325	2020-09-19 02:19:13.414869	10.42.85.10	194.61.24.102	ICMP	60 Echo (ping) reply id=0xef6f, seq=0/0, ttl=128 (request in 84319)
84326	2020-09-19 02:19:13.426119	10.42.85.115	10.42.85.10	TCP	60 50668 → 445 [ACK] Seq=15062 Ack=17270 Win=210092 Len=0

EXHIBIT 7 Supporting packet capture activity around the logon attempts

84336	2020-09-19 02:19:26.469759	194.61.24.102	10.42.85.10	TCP	66 38088 → 3389 [SYN] Seq=0 Win=64256 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250824 TSecr=0 WS=128
84337	2020-09-19 02:19:26.469897	194.61.24.102	10.42.85.10	TCP	66 38088 → 3389 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250824 TSecr=701188
84338	2020-09-19 02:19:26.472244	194.61.24.102	10.42.85.10	TCP	74 38090 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250827 TSecr=0 WS=128
84340	2020-09-19 02:19:26.472579	194.61.24.102	10.42.85.10	TCP	66 38090 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250827 TSecr=701188
84341	2020-09-19 02:19:26.472604	194.61.24.102	10.42.85.10	TCP	74 38092 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250827 TSecr=0 WS=128
84343	2020-09-19 02:19:26.472841	194.61.24.102	10.42.85.10	TCP	66 38092 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250827 TSecr=701188
84344	2020-09-19 02:19:26.495364	194.61.24.102	10.42.85.10	TCP	74 38094 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250850 TSecr=0 WS=128
84345	2020-09-19 02:19:26.495519	194.61.24.102	10.42.85.10	TCP	74 38096 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250850 TSecr=0 WS=128
84348	2020-09-19 02:19:26.495778	194.61.24.102	10.42.85.10	TCP	74 38098 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250850 TSecr=0 WS=128
84349	2020-09-19 02:19:26.495806	194.61.24.102	10.42.85.10	TCP	66 38094 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250850 TSecr=701191
84350	2020-09-19 02:19:26.495823	194.61.24.102	10.42.85.10	TCP	66 38096 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250850 TSecr=701191
84352	2020-09-19 02:19:26.495963	194.61.24.102	10.42.85.10	TCP	74 38100 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976250850 TSecr=0 WS=128
84354	2020-09-19 02:19:26.496123	194.61.24.102	10.42.85.10	TCP	66 38098 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250851 TSecr=701191
84355	2020-09-19 02:19:26.496295	194.61.24.102	10.42.85.10	TCP	66 38100 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976250851 TSecr=701191
84356	2020-09-19 02:19:26.497284	194.61.24.102	10.42.85.10	TLSv1	583 Client Hello
84357	2020-09-19 02:19:26.549024	194.61.24.102	10.42.85.10	TLSv1.2	281 Client Hello
84358	2020-09-19 02:19:26.549093	194.61.24.102	10.42.85.10	SSLv2	120 Client Hello
84359	2020-09-19 02:19:26.549115	194.61.24.102	10.42.85.10	TLSv1.2	124 Client Hello
84360	2020-09-19 02:19:26.549130	194.61.24.102	10.42.85.10	TLSv1.2	162 Client Hello
84361	2020-09-19 02:19:26.549145	194.61.24.102	10.42.85.10	SSL	108 Continuation Data
84371	2020-09-19 02:19:26.792588	194.61.24.102	10.42.85.10	TCP	66 38090 → 3389 [ACK] Seq=216 Ack=852 Win=64128 Len=0 TSval=2976251147 TSecr=701220
84372	2020-09-19 02:19:26.792622	194.61.24.102	10.42.85.10	TCP	66 38100 → 3389 [ACK] Seq=43 Ack=20 Win=64256 Len=0 TSval=2976251147 TSecr=701220
84373	2020-09-19 02:19:26.792649	194.61.24.102	10.42.85.10	TCP	66 38098 → 3389 [ACK] Seq=97 Ack=852 Win=64128 Len=0 TSval=2976251147 TSecr=701220
84375	2020-09-19 02:19:26.792908	194.61.24.102	10.42.85.10	TCP	66 38096 → 3389 [ACK] Seq=59 Ack=852 Win=64128 Len=0 TSval=2976251147 TSecr=701220
84378	2020-09-19 02:19:26.793758	194.61.24.102	10.42.85.10	TCP	74 38102 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976251148 TSecr=0 WS=128
84380	2020-09-19 02:19:26.794136	194.61.24.102	10.42.85.10	TCP	66 38102 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976251149 TSecr=701220
84381	2020-09-19 02:19:26.794224	194.61.24.102	10.42.85.10	TLSv1	583 Client Hello
84383	2020-09-19 02:19:26.795715	194.61.24.102	10.42.85.10	TCP	74 38104 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2976251150 TSecr=0 WS=128

EXHIBIT 8 Sustained RDP (TCP 3389) traffic from the external host in Case001.pcap

04 MALWARE INVESTIGATION

Q6 Was malware used? If so, what was it?

ANSWER Yes — coreupdater.exe**TOOL USED** Volatility, FTK Imager / Autopsy, VirusTotal, Registry Explorer**EVIDENCE** HTTP GET requests from the noted IP for 194.61.24.102/coreupdater.exe, confirmed via file hash on VirusTotal.

Q7.a What process was malicious?

ANSWER coreupdater.exe

```

+ 2385... 15959.468916 10.42.85.10 194.61.24.102 HTTP 291 GET /coreupdater.exe HTTP/1.1
+ 2385... 15959.469636 194.61.24.102 10.42.85.10 HTTP 110 HTTP/1.0 200 OK (application/x-msdos-program)
2658... 16696.836744 10.42.85.115 23.57.58.91 HTTP/X... 1472 POST /fwlink/?LinkId=252669&clcid=0x409 HTTP/1.1
2658... 16696.885023 23.57.58.91 10.42.85.115 HTTP 350 HTTP/1.1 302 Moved Temporarily
2659... 16697.185625 10.42.85.115 20.189.118.208 HTTP/X... 1472 POST /metadata.svc HTTP/1.1
2663... 16697.558423 20.189.118.208 10.42.85.115 HTTP/X... 692 HTTP/1.1 200 OK
2664... 16697.625607 10.42.85.115 23.57.58.91 HTTP/X... 652 POST /fwlink/?LinkId=252669&clcid=0x409 HTTP/1.1
2664... 16697.676328 23.57.58.91 10.42.85.115 HTTP 350 HTTP/1.1 302 Moved Temporarily
2665... 16697.684072 10.42.85.115 20.189.118.208 HTTP/X... 652 POST /metadata.svc HTTP/1.1
2667... 16698.048122 20.189.118.208 10.42.85.115 HTTP/X... 692 HTTP/1.1 200 OK
2667... 16698.103737 10.42.85.115 23.57.58.91 HTTP/X... 1472 POST /fwlink/?LinkId=252669&clcid=0x409 HTTP/1.1

Frame 238565: 291 bytes on wire (2328 bits), 291 bytes captured (2328 bits) on interface 0000 00 0c 29 95 cd 21 00 0c 29 e1 84 e6
Ethernet II, Src: VMware_e1:84:e6 (00:0c:29:e1:84:e6), Dst: VMware_95:cd:21 (00:0c:29:95:cd:21) 0010 01 15 5e 53 40 00 80 06 61 b8 0a 2a
Internet Protocol Version 4, Src: 10.42.85.10, Dst: 194.61.24.102 0020 18 66 f3 ca 00 50 67 70 65 97 5c e8
Transmission Control Protocol, Src Port: 62410, Dst Port: 80, Seq: 1, Ack: 1, Len: 237 0030 04 00 08 97 00 00 47 45 54 20 2f 63
Hypertext Transfer Protocol 0040 70 64 61 74 65 72 2e 65 78 65 20 48
GET /coreupdater.exe HTTP/1.1\r\n 0050 31 2e 31 0d 0a 41 63 63 65 70 74 3a
Accept: */*\r\n 0060 0d 0a 52 65 66 65 72 65 72 3a 20 68
Referer: http://194.61.24.102/\r\n 0070 2f 2f 31 39 34 2e 36 31 2e 32 34 2e
Accept-Encoding: gzip, deflate\r\n 0080 0d 0a 41 63 63 65 70 74 2d 45 6e 63
User-Agent: Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko\r\n 0090 67 3a 20 67 7a 69 70 2c 20 64 65 66
Host: 194.61.24.102\r\n 00a0 0d 0a 55 73 65 72 2d 41 67 65 6e 74
Connection: Keep-Alive\r\n 00b0 7a 69 6c 6c 61 2f 35 2e 30 20 28 57
\r\n 00c0 77 73 20 4e 54 20 36 2e 33 3b 20 57
[Full request URI: http://194.61.24.102/coreupdater.exe] 00d0 3b 20 54 72 69 64 65 6e 74 2f 37 2e
[HTTP request 1/1] 00e0 76 3a 31 31 2e 30 29 20 6c 69 6b 65
[Response in frame: 238574] 00f0 6b 6f 0d 0a 48 6f 73 74 3a 20 31 39

```

EXHIBIT 9 Wireshark HTTP GET request retrieving coreupdater.exe from the attacker host

Wireshark · Export · HTTP object list

Text Filter: 194 Content Type: All Content-Types

Packet	Hostname	Content Type	Size	Filename
236791	194.61.24.102	text/html	228 bytes	\
236809	194.61.24.102	text/html	195 bytes	favicon.ico
238574	194.61.24.102	application/x-msdos-program	7168 bytes	coreupdater.exe
327366	194.61.24.102	text/html	228 bytes	\
339465	194.61.24.102	application/x-msdos-program	7168 bytes	coreupdater.exe

EXHIBIT 10 Exported HTTP objects showing the malicious executable transfer

10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfdad

61/70 security vendors flagged this file as malicious

Reanalyze Similar More

10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfdad

coreupdater.exe

Size 7.00 KB Last Analysis Date 1 month ago

Community Score -12

peexe idle spreader runtime-modules 64bits assembly direct-cpu-clock-access

EXE

EXHIBIT 11 VirusTotal: 61 of 70 vendors flagged the sample as malicious

Q7.b What IP address delivered the payload?

ANSWER 194.61.24.102

TOOL USED Wireshark

2385... 15959.468916 10.42.85.10 194.61.24.102 HTTP 291 GET /coreupdater.exe HTTP/1.1

EXHIBIT 12 Packet showing the GET request for the payload from the delivery host

Q7.c What IP address is the malware calling to?

Answer: The malware initiated outbound communication with the IP address 203.78.103.109, identified through Wireshark analysis of the Case001.pcap file. This activity occurred shortly after the system downloaded coreupdater.exe, suggesting that this IP served as a Command-and-Control (C2) server.

I filtered HTTP and TCP traffic in **Wireshark** (Wireshark Foundation, 2024) and correlated timestamps of the download with subsequent external connections. VirusTotal confirmed that this IP has a reputation for hosting malicious content, reinforcing its role as a C2 server.

Outbound communication to a C2 server typically indicates that the malware is attempting to exfiltrate data or receive further instructions from the attacker (MITRE ATT&CK: T1071.001).

TOOL USED Wireshark, VirusTotal

203.78.103.109 7 / 94 18362 TH

EXHIBIT 13 VirusTotal reputation result for the C2 address

2422...	16031.095815	10.42.85.10	203.78.103.109	TCP	60 62414 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
2422...	16031.281664	203.78.103.109	10.42.85.10	TCP	58 443 → 62414 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=4
2422...	16031.282793	203.78.103.109	10.42.85.10	SSLv2	1514 Encrypted Data, Encrypted Data, Encrypted Data, Enc
2422...	16031.282819	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282836	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282852	203.78.103.109	10.42.85.10	SSLv2	1514 Encrypted Data, Encrypted Data
2422...	16031.282869	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282884	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282900	203.78.103.109	10.42.85.10	SSLv2	1514 Encrypted Data
2422...	16031.282916	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282932	203.78.103.109	10.42.85.10	TCP	1514 [TCP segment of a reassembled PDU]
2422...	16031.282963	10.42.85.10	203.78.103.109	TCP	60 62414 → 443 [ACK] Seq=1 Ack=4385 Win=65536 Len=0
2422...	16031.283095	10.42.85.10	203.78.103.109	TCP	60 62414 → 443 [ACK] Seq=1 Ack=13145 Win=65536 Len=0

EXHIBIT 14 Encrypted outbound sessions to the C2 host over TCP 443

Q7.d Where is the malware on disk?

ANSWER C:\Windows\System32\coreupdater.exe

TOOL USED Volatility

```

\Windows\System32\coreupdater.exereupdater.exe.2424urv.partial
\Windows\System32\coreupdater.exereupdater.exe
\Windows\System32\coreupdater.exereupdater.exe.2424urv.partial
\Windows\System32\coreupdater.exereupdater.exe

```

EXHIBIT 15 Path references to the executable recovered from memory

Q7.e When did it first appear?**ANSWER** 2020-09-19 03:24:12 UTC**TOOL USED** FTK Imager / Autopsy

Parent Path	File Name	Extension	Is Directory	Has Ads	Is Ads	File Size	Created0x10	Created0x30	Last Modified0x10	Last Modified0x30
.\Windows\System32	coreupdater.exe	.exe				7168	2020-09-19 03:24:12		2020-09-19 03:24:06	

EXHIBIT 16 File system metadata showing the creation timestamp of the executable**Q7.f Did someone move it?**

Yes. It was moved from the Administrator's Downloads folder to the C drive of the DC and desktop systems. Later entries for the file indicated the parent entry number to be that of System32.

2020-09-19 03:24:12	coreupdater.exe	.exe	87137	FileCreate	1	84880
2020-09-19 03:24:12	coreupdater.exe	.exe	87137	FileCreate Close	1	84880
2020-09-19 03:24:12	coreupdater.exe	.exe	87137	FileDelete Close	1	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	FileCreate	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	FileCreate Close	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataTruncation	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataTruncation Close	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataExtend	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataOverwrite DataExtend	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataOverwrite DataExtend Basic...	2	84880
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	DataOverwrite DataExtend Basic...	2	84880
2020-09-19 03:24:12	coreupdater[1].exe	.exe	76711	FileDelete Close	12	87050
2020-09-19 03:24:12	coreupdater.exe.2424urv.partial	.partial	87137	RenameOldName	2	84880
2020-09-19 03:24:12	coreupdater.exe	.exe	87137	RenameNewName	2	84880
2020-09-19 03:24:12	coreupdater.exe	.exe	87137	RenameNewName Close	2	84880
2020-09-19 03:24:50	coreupdater.exe	.exe	87137	RenameOldName	2	84880

EXHIBIT 17 File create, rename and data-overwrite events for the executable

16019	2020-09-17 16:46:26	Downloads	84880	ObjectIdChange Close	1
16018	2020-09-17 16:46:26	Downloads	84880	ObjectIdChange	1
15873	2020-09-17 16:46:25	Downloads	84880	BasicInfoChange Close	1
15872	2020-09-17 16:46:25	Downloads	84880	BasicInfoChange	1
14883	2020-09-17 16:46:16	Downloads	84880	BasicInfoChange Close	1
14882	2020-09-17 16:46:16	Downloads	84880	BasicInfoChange	1
14881	2020-09-17 16:46:16	Downloads	84880	BasicInfoChange Close	1
14880	2020-09-17 16:46:16	Downloads	84880	BasicInfoChange	1
14182	2020-09-17 16:46:15	Downloads	84880	SecurityChange BasicInfoChange...	1
14181	2020-09-17 16:46:15	Downloads	84880	SecurityChange BasicInfoChange	1
14180	2020-09-17 16:46:15	Downloads	84880	SecurityChange	1
14179	2020-09-17 16:46:15	Downloads	84880	FileCreate Close	1
14178	2020-09-17 16:46:15	Downloads	84880	FileCreate	1

EXHIBIT 18 Corresponding activity within the Downloads folder

2020-09-19 03:40:42	coreupdater.exe	.exe	26213	RenameNewName	4	3254
2020-09-19 03:40:42	coreupdater.exe	.exe	26213	RenameNewName Close	4	3254
2020-09-19 03:40:42	coreupdater.exe	.exe	26213	SecurityChange	4	3254
2020-09-19 03:40:42	coreupdater.exe	.exe	26213	SecurityChange Clo...	4	3254
2020-09-19 03:40:49	coreupdater.exe	.exe	26213	StreamChange	4	3254
2020-09-19 03:40:49	coreupdater.exe	.exe	26213	StreamChange Close	4	3254

EXHIBIT 19 Subsequent rename and stream-change events

35946	2020-09-19 03:40:42	System32	3254	ObjectIdChange
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EXHIBIT 20 Parent entry resolving to System32, confirming the relocation**Q7.g What were the capabilities of this malware?**

This tool has a vast array of capabilities, as it provides the attacker with a shell to the victim machine. From here they are able to further explore the victim's device, pivot to other devices in the network, or run / execute other payloads and commands.

TOOL USED VirusTotal, Hybrid Analysis (Joe Sandbox report)

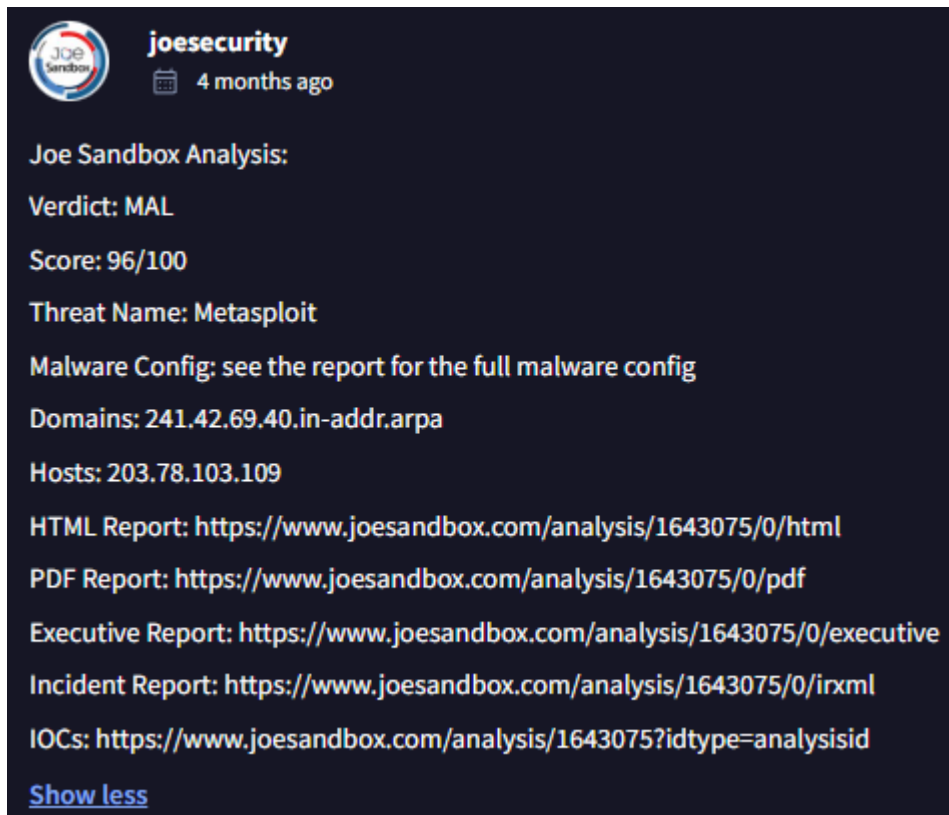


EXHIBIT 21 Sandbox verdict: malicious, score 96/100, threat name Metasploit

Q7.h Is this malware easily obtained?

ANSWER	Yes — Metasploit is an open-source red teaming tool, so it is safe to say it is pretty accessible.
TOOL USED	VirusTotal, Metasploit, Hybrid Analysis

Q7.i Was this malware installed with persistence on any machine?

Answer: Yes. Persistence was established on the DC01 system via a malicious Windows service named `coreupdater`, configured to run automatically at system startup.

Using **FTK Imager / Autopsy**, I exported the SYSTEM registry hive and analyzed it using **Registry Explorer** (Zimmerman Tools, 2023). Under `HKLM\SYSTEM\CurrentControlSet\Services`, I identified a new service created at 03:27:49 UTC — shortly after the malware was dropped to disk. The service name matched the malicious executable, and its start type was set to "auto", ensuring execution on reboot.

I also searched for **Event ID 7045** in the extracted `System.evtx` logs using **Event Log Explorer**, confirming that a new service was installed. This aligns with common persistence techniques documented in MITRE ATT&CK under T1543.003 — Windows Service.

TOOL USED	FTK Imager (registry), Event Log Explorer
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Q7.j When and where was persistence set?

The malware created a service to run on startup on the DC as well as a registry key on both systems. It first occurred on the DC at 03:27:49.

CITADEL-DC01.C137.local	2020-09-19 04:39:59	7045	Name: AccessData Driver	StartType: demand start
CITADEL-DC01.C137.local	2020-09-19 03:56:55	7045	Name: pmhrio	StartType: demand start
CITADEL-DC01.C137.local	2020-09-19 03:44:29	7045	Name: mszhao	StartType: demand start
CITADEL-DC01.C137.local	2020-09-19 03:27:49	7045	Name: coreupdater	StartType: auto start
CITADEL-DC01.C137.local	2020-09-19 03:25:44	7045	Name: outngo	StartType: demand start

EXHIBIT 22 Event ID 7045 service installation records, including coreupdater set to auto start

System\CurrentControlSet\Services\coreupdater enabled Services System-wide coreupdater: c:\windows\system32\coreupdater.exe

EXHIBIT 23 Registry service entry pointing to the malicious executable in System32

05 INDICATORS OF COMPROMISE

The following indicators were consolidated from the findings above and can be used for blocking, hunting and scoping across the wider environment.

TYPE	INDICATOR	CONTEXT
IPv4	194.61.24.102	RDP brute-force source and payload delivery host
IPv4	203.78.103.109	Command-and-control server (outbound TCP 443)
Filename	coreupdater.exe	Malicious executable, 7,168 bytes
SHA-256	10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6	Flagged by 61/70 vendors on VirusTotal
File path	C:\Windows\System32\coreupdater.exe	Final location after relocation from Downloads
Service	coreupdater	Windows service, start type "auto"
Registry	HKLM\SYSTEM\CurrentControlSet\Services\coreupdater	Persistence mechanism
Domain	241.42.69.40.in-addr.arpa	Observed in sandbox analysis of the sample
URL	http://194.61.24.102/coreupdater.exe	Payload download over HTTP

Detection references

ARTIFACT	DETAIL
Event ID 4625	Failed logon attempts — RDP brute force
Event ID 4624	Successful logon following the brute force

ARTIFACT	DETAIL
Event ID 7045	New service installed (persistence)
T1071.001	Application Layer Protocol: Web Protocols (C2 channel)
T1543.003	Create or Modify System Process: Windows Service (persistence)

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